55. High-Speed Analog Comparator (ACMPHS)

55.1 Overview

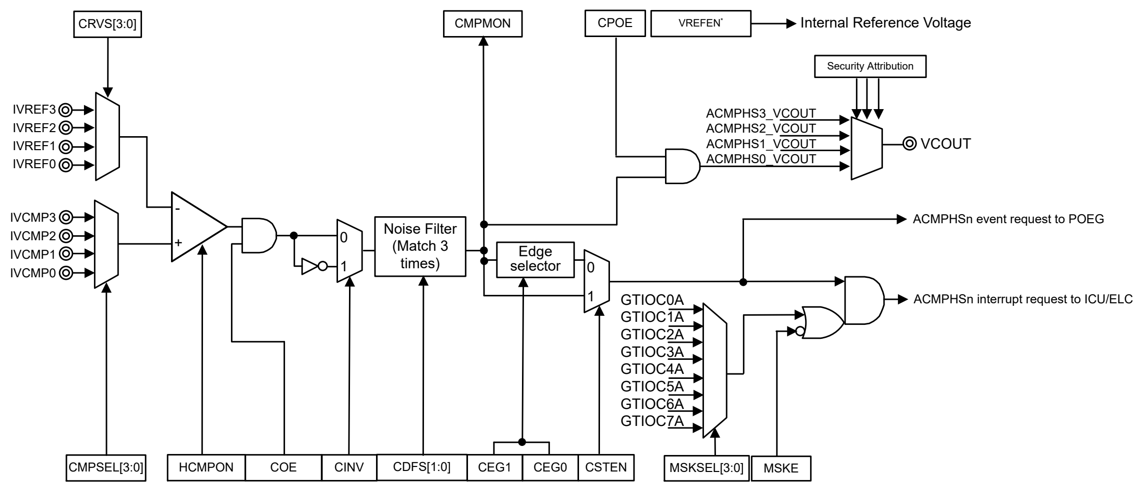
The High-Speed Analog Comparator (ACMPHS) can be used to compare an analog input voltage with a reference voltage and to provide a digital output based on the result of conversion. Both the analog input voltage and the reference voltage can be provided to the ACMPHS from internal sources (D/A converter output or internal reference voltage) and an external source. Such flexibility is useful in applications that require go/no-go comparisons to be performed between analog signals without necessarily requiring A/D conversion.

Table 55.1 lists the ACMPHS specifications, Figure 55.1 shows a block diagram, and Table 55.2 shows the input source configurations.

Table 55.1 ACMPHS specifications

Parameter	Specifications
Number of channels	4 channels: ACMPHSn (n = 0 to 3)
Analog input voltage	● Output from internal D/A converter ● Input from an external source (compatible with internal A/D converter input pin (one selectable))
Reference voltage	● Internal reference voltage (Vref) ● Output from internal D/A converter ● Input from an external source (compatible with internal A/D converter input pin (one selectable))
ACMPHS output	● Comparison result output to terminal PIN ● Generation of ELC event output ● Monitor output from register ● Generation of interrupt request output
Interrupt request signal	● Interrupt request generated on valid edge detection from comparison result (when CMPCTL.CSTEN = 0*1) ● Selectable to rising edge, falling edge, or both edges (when CMPCTL.CSTEN = 0*1) ● Rising edge only (when CMPCTL.CSTEN = 1*1)
Digital filter function	● Selectable to one of three sampling frequencies ● Not using the filter function is selectable
Module-stop function	Module-stop state can be set for each groups to reduce power consumption.
TrustZone Filter	Security and Privilege attribution can be set for each group

Note 1. Interrupt request signal selection restriction detail refers to section 55.5. ACMPHS Interrupts and section 55.6. ACMPHS Output to the Event Link Controller (ELC).



n = 0 to 3

CMPSEL[3:0] : bits in ACMPHSn.CMPSEL0 register
CRVS[3:0] : bits in ACMPHSn.CMPSEL1 register
CPOE, VREFEN : bits in ACMPHSn.CPIOC register
CMPMON : bits in ACMPHSn.CMPMON register
HCMPON, COE, CINV, CDFS[1:0], CEG1, CEG0, CSTEN : bits in ACMPHSn.CMPCTL register
MSKSEL[3:0] : bits in ACMPHSn.CMPSKCTL register
MSEK : bits in ACMPHSn.CMPINTCTL register

Note: VREFEN exists only in ACMPHS0.

Figure 55.1 ACMPHS block diagram

Table 55.2 Input source configuration of the ACMPHS

Comparator	Reference voltage input source				Analog voltage input source				Output pin
	IVREF3	IVREF2	IVREF1	IVREF0	IVCMP3	IVCMP2	IVCMP1	IVCMP0	
ACMPHS0	DA0	Vref	AN009	AN008	AN001	AN000	DA1	AN017	VCOUT*1
ACMPHS1	DA0	Vref	AN009	AN008	AN003	AN002	DA1	AN016	VCOUT*1
ACMPHS2	DA0	Vref	AN009	AN008	AN005	AN004	DA1	AN015	VCOUT*1
ACMPHS3	DA0	Vref	AN009	AN008	AN007	AN006	DA1	AN014	VCOUT*1

Note 1. ACMPHS0 to ACMPHS3 compare outputs are bundled with the VCOUT pin.

55.2 Register Descriptions

55.2.1 CMPCTL : Comparator Control Register

Base address: ACMPHSn = 0x4023_6000 + 0x0100 × n (n = 0 to 3)
ACMPHSn_NS = 0x5023_6000 + 0x0100 × n (n = 0 to 3)

Offset address: 0x000

Bit position:	7	6	5	4	3	2	1	0
Bit field:	HCMPON	CDFS[1:0]		CEG[1:0]		CSTEN	COE	CINV
Value after reset:	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	CINV	Comparator Output Polarity Selection*1 *2 0: Do not invert comparator output 1: Invert comparator output	R/W

Bit	Symbol	Function	R/W
1	COE	Comparator Output Enable 0: Disable comparator output (output signal is low level) 1: Enable comparator output	R/W
2	CSTEN	Interrupt Select*3 0: Output through the edge selector 1: Output directly	R/W
4:3	CEG[1:0]	Selection of Valid Edge (Edge Selector) 0 0: Do not detect edge 0 1: Detect rising edge 1 0: Detect falling edge 1 1: Detect both edges	R/W
6:5	CDFS[1:0]	Noise Filter Selection *1 *2 *3 *4 0 0: Do not use noise filter 0 1: Use noise filter sampling frequency of PCLKB/2^3 1 0: Use noise filter sampling frequency of PCLKB/2^4 1 1: Use noise filter sampling frequency of PCLKB/2^5	R/W
7	HCMPON	Comparator Operation Control*5 0: Stop operation (comparator outputs a low-level signal) 1: Enable operation (enables input to the comparator pins)	R/W

Note: S-TYPE-3, P-TYPE-3

Note 1. Disable the ACMPHS output (COE = 0) before changing the CDFS[1:0] and CINV bits.

Note 2. If the CDFS[1:0] and CINV bits are changed, an ACMPHS interrupt request and an ELC event might be generated. Before changing these bits, set the ELSRn register to 0 (the ACMPHS output is not linked). After changing these bits, clear the IR flag in the IELSRn register to 0 to clear the interrupt status.

Note 3. Set the CSTEN bit to 1 and the CDFS[1:0] bits to 00b if the ACMPHS interrupt causes release of Software Standby mode. CSTEN is supported only by the ACMPHS0. ACMPHSn.CMPCTL.CTESN (n = 1 to 3) must be set to 0.

Note 4. If the CDFS[1:0] bits are changed from 00b (noise filter not used) to a value other than 00b (noise filter used), perform sampling four times and update the filter output, and then use the ACMPHS interrupt request or the ELC event.

Note 5. A stabilization wait time is required to permit ACMPHS operation after enabling it (HCMPON = 1). The operation stabilization wait time for ACMPHS modules 0 to 1 is 300 ns.

Note: Set this register before setting registers in the POEG when using comparator output as a POEG source.

The CMPCTL register controls the ACMPHS operation, enables or disables the ACMPHS output, selects the noise filter, selects the valid edge of the interrupt signal, and selects the interrupt.

55.2.2 CMPSEL0 : Comparator Input Select Register

Base address: ACMPHSn = 0x4023_6000 + 0x0100 × n (n = 0 to 3)
ACMPHSn_NS = 0x5023_6000 + 0x0100 × n (n = 0 to 3)

Offset address: 0x004

Bit position:	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	CMPSEL[3:0]			
Value after reset:	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
3:0	CMPSEL[3:0]	Comparator Input Selection*1 *2 *3 0x00: Do not input 0x01: Select IVCMP0 0x02: Select IVCMP1 0x04: Select IVCMP2 0x08: Select IVCMP3 Others: Setting prohibited	R/W
7:4	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

Note 1. Use the following procedure to change the CMPSEL[3:0] bits. Writing a value other than 0x00 while the value of the CMPSEL0 register is not 0x00 is invalid. Writing 1 to two or more bits is also invalid. In both cases, the previous value is retained.

To change the CMPSEL[3:0] bits:
1. Set the CMPCTL.COE bit to 0.
2. Set the CMPSEL0 register to 0x00.
3. Set a new value in the CMPSEL[3:0] bits, with 1 set in only one of the bits.
4. Wait for the input switching stabilization wait time (200 ns).
5. Set the CMPCTL.COE bit to 1.
6. Clear the IR flag in the IELSRn register to clear the interrupt status.

Note 2. For details, see Table 55.2.
Note 3. If ACMPHSn level detection signal is used as a POEG source, write access to these bits after the setting of any register in the POEG may generate a POEG source.

55.2.3 CMPSEL1 : Comparator Reference Voltage Select Register

Base address: ACMPHSn = 0x4023_6000 + 0x0100 × n (n = 0 to 3)
ACMPHSn_NS = 0x5023_6000 + 0x0100 × n (n = 0 to 3)

Offset address: 0x008

Bit position:	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	CRVS[3:0]			
Value after reset:	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
3:0	CRVS[3:0]	Reference Voltage Selection*1*2*3*4 0x00: Do not input 0x01: Select IVREF0 0x02: Select IVREF1 0x04: Select IVREF2 0x08: Select IVREF3 Others: Setting prohibited	R/W
7:4	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3
Note 1. Use the following procedure to change the CRVS[3:0] bits. Writing a value other than 0x00 while the value of the CMPSEL1 register is not 0x00 is invalid. Writing 1 to two or more bits is also invalid. In both cases, the previous value is retained.
To change the CRVS[3:0] bits:
1. Set the CMPCTL.COE bit to 0.
2. Set the CMPSEL1 register to 0x00.
3. Set a new value to the CRVS[3:0] bits, with 1 set in only one of the bits.
4. Wait for the input switching stabilization wait time (200 ns)
5. Set the CMPCTL.COE bit to 1.
6. Clear the IR flag in the IELSRn register to clear the interrupt status.

Note 2. For details, see Table 55.2.
Note 3. When the on-chip D/A converter output voltage is used, set the D/A converter to generate comparator reference voltage before enabling comparator operation (CMPCTL.HCMPON = 1). For details on setting the D/A converter, see section 53, 12-Bit D/A Converter (DAC12).
Note 4. If ACMPHSn level detection signal is used as a POEG source, write access to these bits after the setting of any register in the POEG may generate a POEG source.

55.2.4 CMPMON : Comparator Output Monitor Register

Base address: ACMPHSn = 0x4023_6000 + 0x0100 × n (n = 0 to 3)
ACMPHSn_NS = 0x5023_6000 + 0x0100 × n (n = 0 to 3)

Offset address: 0x00C

Bit position:	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	COMP MON
Value after reset:	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	COMPMON	Comparator Output Monitor*1 0: Comparator output is low 1: Comparator output is high	R
7:1	—	These bits are read as 0. The write value should be 0.	R

Note: S-TYPE-3, P-TYPE-3

Note 1. When ACMPHS operation is enabled (CMPCTL.HCMPON = 1 and CMPCTL.COE = 1) but the noise filter is not in use (CDFS[1:0] = 00b), design the software so that the COMPMON bit is read twice and the values are only used after the two consecutive values match.

55.2.5 CPIOC : Comparator Output Control Register

Base address: ACMPHSn = 0x4023_6000 + 0x0100 × n (n = 0 to 3)
ACMPHSn_NS = 0x5023_6000 + 0x0100 × n (n = 0 to 3)

Offset address: 0x010

Bit position:	7	6	5	4	3	2	1	0
Bit field:	VREF EN	—	—	—	—	—	—	CPOE
Value after reset:	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	CPOE	External Pin Output Enable Comparison result by the comparator is output to an external pin. 0: Output to the comparator external pin is disabled (the output signal is fixed to low) 1: Output to the comparator external pin is enabled	R/W
6:1	—	These bits are read as 0. The write value should be 0.	R/W
7	VREFEN	Internal Vref Enable*1 0: Disable internal Vref 1: Enable internal Vref	R/W

Note: S-TYPE-3, P-TYPE-3

Note 1. For ACMPHS modules 0 to 3, VREFEN exists only in ACMPHS0.CPIOC. When using the internal Vref in ACMPHSn.CMPCTL.CTESN (n = 0 to 3), set the VREFEN bit in ACMPHS0.CPIOC to 1. Bit [7] in ACMPHSn.CPIOC (n = 1 to 3) should be 0 regardless of whether or not the internal Vref is used.

55.2.6 CPINTCTL : Comparator Interrupt Control Register

Base address: ACMPHSn = 0x4023_6000 + 0x0100 × n (n = 0 to 3)
ACMPHSn_NS = 0x5023_6000 + 0x0100 × n (n = 0 to 3)

Offset address: 0x040

Bit position:	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	MSKE
Value after reset:	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	MSKE	Comparator Interrupt Periodic Mask Enable 0: Disable interrupt masking (Default) 1: Enable interrupt masking by GPT output signal selected by CPMSKCTL.MSKSEL [2:0]	R/W
7:1	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

This register controls the interrupt to ICU/ELC, while the event to POEG (Port Output Enable) is not controlled by this register.

55.2.7 CPMSKCTL : Comparator Interrupt Mask Control Register

Base address: ACMPHSn = 0x4023_6000 + 0x0100 × n (n = 0 to 3)
ACMPHSn_NS = 0x5023_6000 + 0x0100 × n (n = 0 to 3)

Offset address: 0x044

Bit position:	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	MSKSEL[2:0]		
Value after reset:	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
2:0	MSKSEL[2:0]	Comparator Interrupt Periodic Mask Selection 0 0 0: Enable interrupt masking by GTIOC0A output signal 0 0 1: Enable interrupt masking by GTIOC1A output signal 0 1 0: Enable interrupt masking by GTIOC2A output signal 0 1 1: Enable interrupt masking by GTIOC3A output signal 1 0 0: Enable interrupt masking by GTIOC4A output signal 1 0 1: Enable interrupt masking by GTIOC5A output signal 1 1 0: Enable interrupt masking by GTIOC6A output signal 1 1 1: Enable interrupt masking by GTIOC7A output signal	R/W
7:3	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

This register controls the interrupt to ICU/ELC, while the event to POEG(Port Output Enable) is not controlled by this register.

55.3 Operation

The ACMPHS compares a reference voltage to an analog input voltage. Operation is not guaranteed when the values of registers are changed during ACMPHS operation. Table 55.3 shows the procedures for setting the registers associated with ACMPHS.

Table 55.3 Procedure for setting registers associated with ACMPHSn (n = 0 to 3) (1 of 2)

Step	Register	Bit	Setting
1	Associated MSTPCRD register	MSTPD28 to MSTPD27	0: Input clock supply.
2	Associated pin function control register (PFS)	ASEL	1: Select the function of pins IVREF and IVCMP.
3	ACMPHSn.CPIOC	VREFEN	1: When using the internal Vref.
4	Associated D/A convertor		When using the D/A convertor, select in the register.
5	CMPSEL0, CMPSEL1	CMPSEL[3:0] CRVS[3:0]	Select the ACMPHSn input, with 1 set in only one of the bits.
6	CMPCTL	CDFS[1:0], CEG[1:0], and CINV	Set up ACMPHSn control.
		HCMPON	1: Enable ACMPHSn operation.
7	CMPSKCTL	MSKCTL[2:0]	Select the interrupt mask source signal (from GPT)
8	CPINT	MSKE	1: Enable the interrupt mask function if needed
9	Waiting for the ACMPHS stabilization time (minimum 300 ns).		
10	CMPCTL	COE	1: Enable ACMPHSn output.
11	CPIOC	CPOE	Set the VCOUT output
	Associated pin function control register (PFS)	PSEL, PMR	Select the VCOUT port function.
12	IELSRn	IR, IELS[8:0]	When using an interrupt, select the interrupt status flag and the ICU event link.*1
13	ELSRn	ELS[8:0]	When using an ELC, select the event link*2.

Table 55.3 Procedure for setting registers associated with ACMPHSn (n = 0 to 3) (2 of 2)

Step	Register	Bit	Setting
14	Operation started		
15	CMPCTL	COE	0: When changing IVREF or IVCMP, to disable ACMPHSn output.
16	CMPSEL1	CRVS[3:0]	Change the CMPSEL1 bits as follows: 1. Set bits CMPSEL1 to 0000 0000b. 2. Set a new value to the CMPSEL1 bits, with 1 set in only one of the bits.
	CMPSEL0	CMPSEL[3:0]	Change the CMPSEL0 bits as follows: 1. Set bits CMPSEL0 to 0000 0000b. 2. Set a new value to the CMPSEL0 bits, with 1 set in only one of the bits.
17	Waiting for the ACMPHS switching stabilization time (minimum 200 ns).		
18	CMPCTL	COE	1: Enable ACMPHSn output.
19	Operation restarted		

Note 1. After ACMPHSn is set, an unnecessary interrupt might occur until operation becomes stable, so initialize the interrupt flag.
Note 2. After ACMPHSn is set, an unnecessary interrupt might occur until operation becomes stable, so initialize the event link select.

Figure 55.2 and Figure 55.3 show an example of ACMPHS operation. The VCOUT output becomes 1 when the analog input voltage is higher than the ACMPHS reference input voltage, and the VCOUT output becomes 0 when the analog input voltage is lower than the reference voltage. When the ACMPHS output changes, an interrupt request and an ELC event are output.

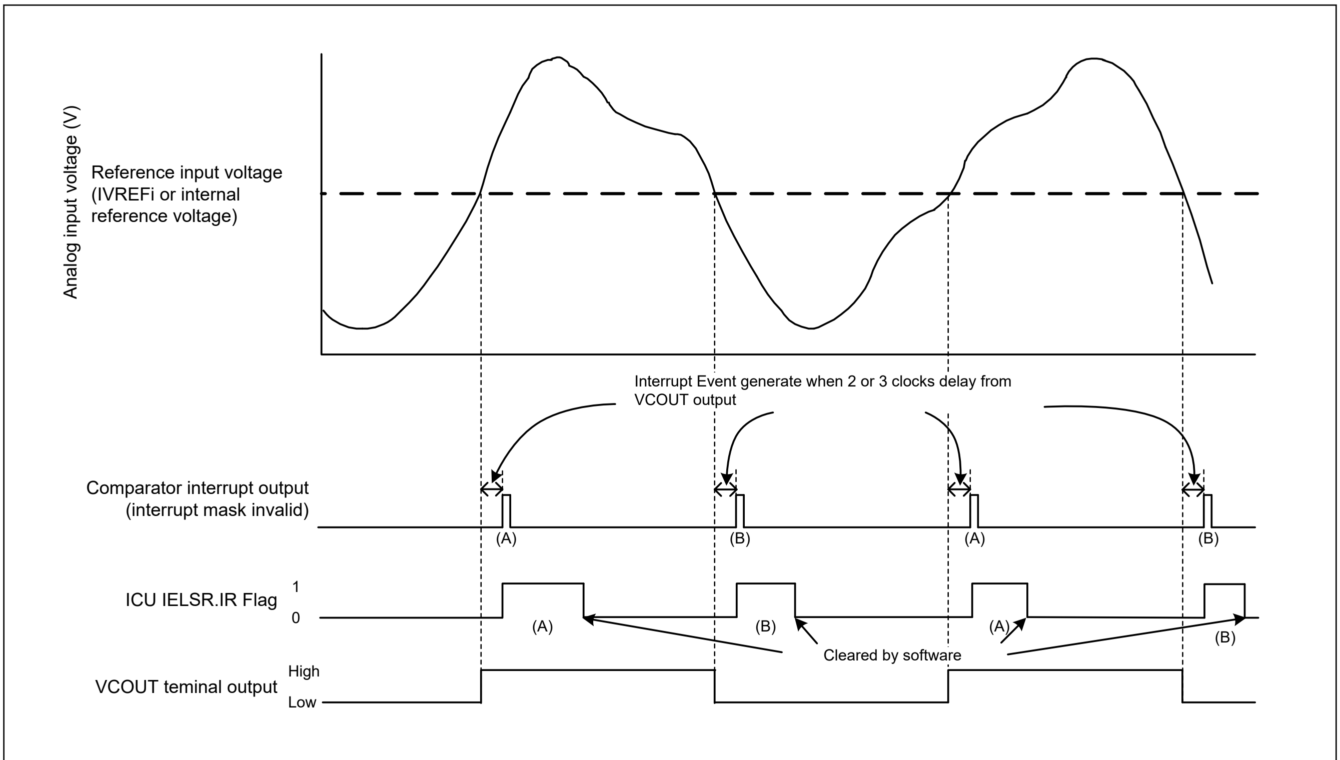


Figure 55.2 ACMPHS operation example (interrupt mask function invalid)

Figure 55.2 applies when CPOE = 1 (pin output enabled), CDFS[1:0] = 00b (filter not used), and CEG[1:0] = 11b (both-edge detection selected). When CINV = 0, CEG[1:0] = 01b (rising-edge detection selected for non-inversion output signal from the ACMPHS), the IELSR.IR flag changes as shown by (A) only. When CINV = 0, CEG[1:0] = 10b (falling-edge detection selected for non-inversion output signal from the ACMPHS), the IR flag changes as shown by (B) only. When CPOE = 1, VCOUT directly outputs.

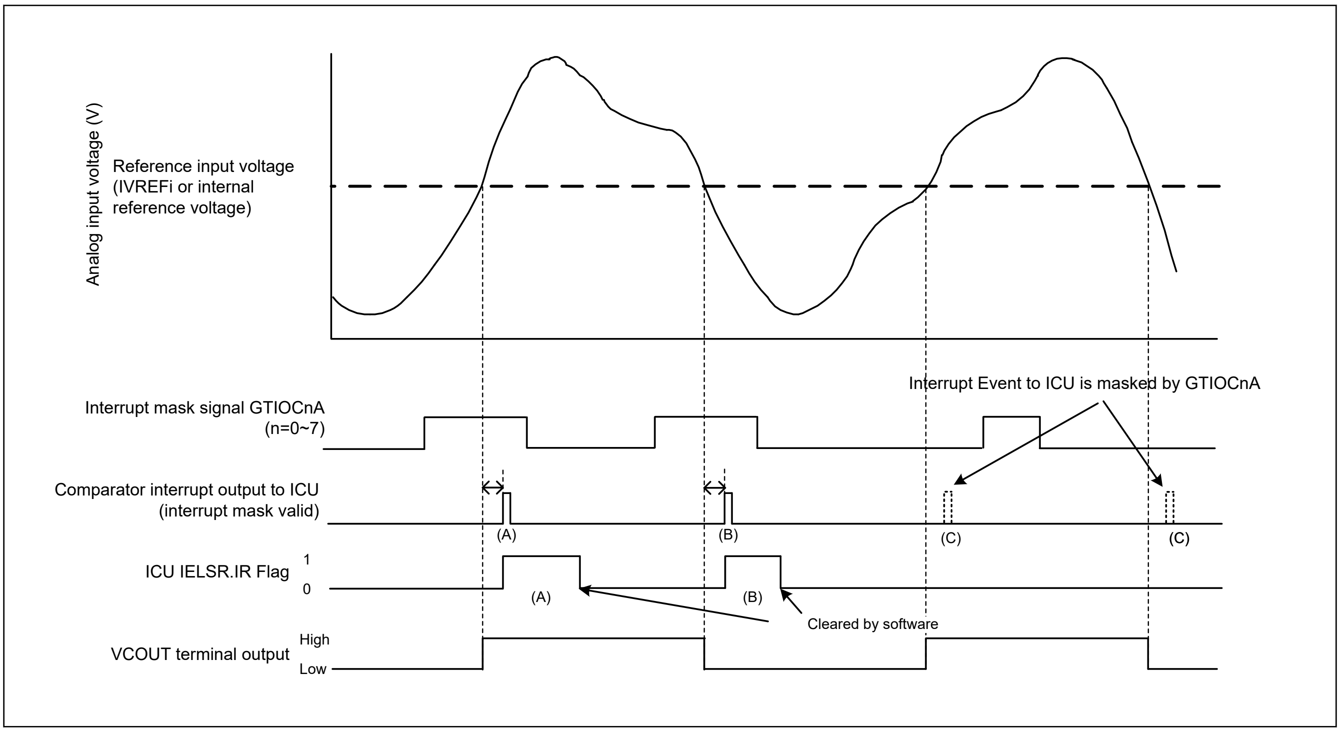


Figure 55.3 ACMPHS operation example (interrupt mask function valid)

Figure 55.3 applies when CPOE = 1 (pin output enabled), CDFS[1:0] = 00b (filter not used), CEG[1:0] = 11b (both-edge detection selected). When CINV = 0, CEG[1:0] = 01b (rising-edge detection selected for non-inversion output signal from the ACMPHS), the IELSR.IR flag changes as shown by (A) only. When CINV = 0, CEG[1:0] = 10b (falling-edge detection selected for non-inversion output signal from the ACMPHS), the IR flag changes as shown by (B) only.

When MSKE = 1 and MSKSEL[2:0] = 000b, the interrupt signal output to ICU is masked by GTIOC0A when GTIOC0A = low level as shown by (C), while it passes through when GTIOC0A = high level.

When CPOE = 1, VCOUT directly outputs, regardless of the MSKE setting.

55.4 Noise Filter

The ACMPHS contains a noise filter. The sampling clock can be selected in the CMPCTL.CDFS[1:0] bits. The comparator output signal is sampled every sampling clock, and if the same value is sampled three times, the noise filter output at the next sampling clock cycle is used as the ACMPHS output.

Figure 55.4 shows the configuration of the noise filter and edge detector, and Figure 55.5 shows an example of noise filter and interrupt operation.

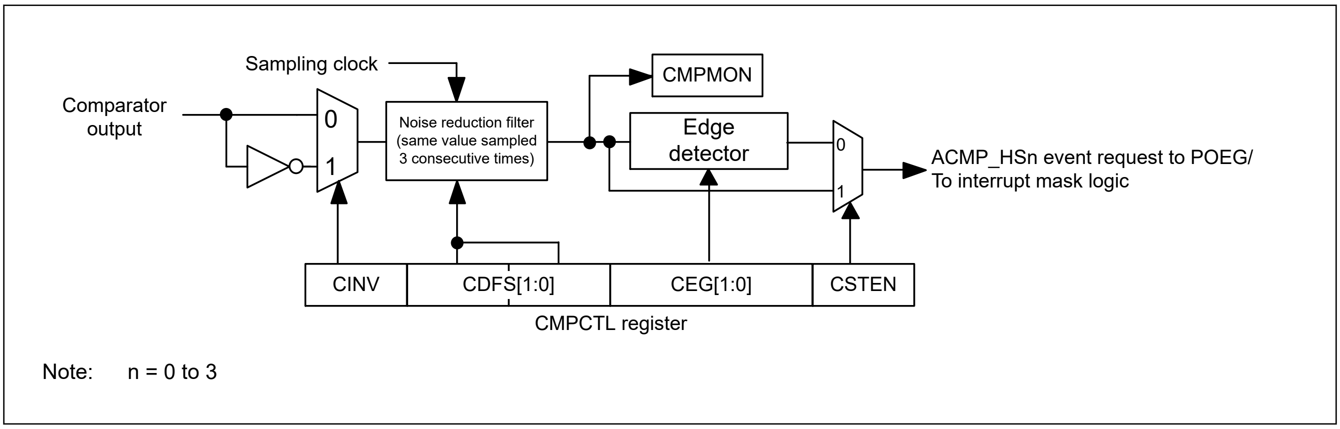


Figure 55.4 Noise filter and edge detection configuration

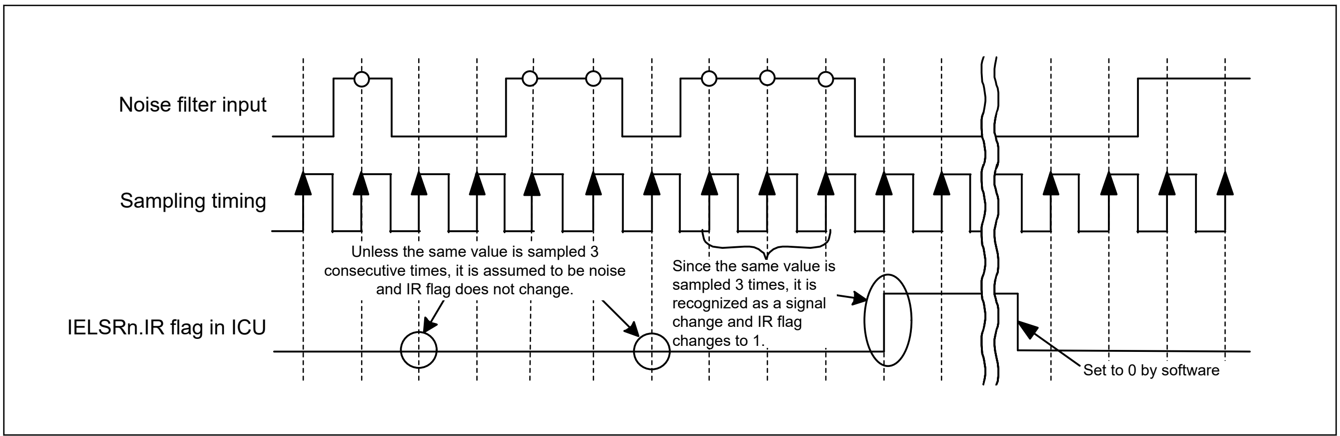


Figure 55.5 Noise filter and interrupt operation example

The operation example in Figure 55.5 applies when the CMPCTL.CDFS[1:0] bits are 01b, 10b, or 11b (noise filter used).

55.5 ACMPHS Interrupts

The ACMPHS generates four interrupt requests from sources ACMPHSn (n = 0 to 3). To use an ACMPHS interrupt, select it in the IELSR register in the Interrupt Controller Unit (ICU). Select the interrupt request in the CMPCTL.CSTEN bit either through the edge selector, or not.

Interrupt event to ICU/ELC can be masked by GPT output (GTIOCnA (n = 0 to 7)), it's controlled by CPINTCTL.MSKE bit and CPMSKCTL.MSKSEL[2:0] bits. Detail refer to Figure 55.3.

Event to POEG can't be masked by GPT output (GTIOCnA (n = 0 to 7)), it's not controlled by CPINTCTL.MSKE bit and CPMSKCTL.MSKSEL[2:0] bits.

When using the ACMPHS interrupt through the edge selector, set at least one of the CMPCTL.CEG[1:0] bits to 1 (to a value other than 00b for no edge selection). Set the CMPCTL.CSTEN bit to 0 (output through the edge selector) in Normal mode, CPU Sleep mode, and CPU Deep Sleep mode.

To use the ACMPHS interrupt in Software Standby mode, set the CMPCTL.CSTEN bit to 1 (direct output), set the CMPCTL.CDFS[1:0] bit to 00b (digital noise filter not used), and set CMPCTL.CINV as follows:

- When detecting compare result 0 to 1, set CMPCTL.CINV to 0 (comparator output not inverted)
- When detecting compare result 1 to 0, set CMPCTL.CINV to 1 (comparator output inverted).

An ACMPHS0 interrupt request can be used to release Software Standby mode. ACMPHS1, ACMPHS2, and ACMPHS3 cannot be used.

ACMPHSn (n = 0 to 3) cannot be used in Deep Software Standby mode.

For details on the register settings related to ACMPHS interrupt requests, see section 55.2.1. CMPCTL : Comparator Control Register, section 55.2.6. CPINTCTL : Comparator Interrupt Control Register and section 55.2.7. CPMSKCTL : Comparator Interrupt Mask Control Register.

55.6 ACMPHS Output to the Event Link Controller (ELC)

The ELC uses the ACMPHS interrupt request signal as an ELC event signal, enabling link operation for the preset module. To use the ACMPHS ELC event, select them in the ELSR register in the ELC. When using the ELC event request, set the CMPCTL.CSTEN bit to 0 (output through the edge selector). Also set at least one of the CMPCTL.CEG[1:0] bits to 1 (to a value other than 00b for no edge selection).

Interrupt event to ELC can be masked by GPT output (GTIOCnA (n = 0 to 7)), it's controlled by CPINTCTL.MSKE bit and CPMSKCTL.MSKSEL[2:0] bits.

55.7 ACMPHS Pin Output

The comparison result from the ACMPHS can be output to external pins. Use the CMPCTL.CINV and CPIOC.CPOE bits to set the output polarity (non-inverted or inverted output) and enable or disable output. To output the ACMPHS comparison result to the VCOUT output pin, set the associated port mn pin function control register (PmnPFS) in the I/O register.

Only when the security attribution of Port and ACMPHS match, the ACMPHS comparison result is output to the VCOUT pin as shown in the Table 55.4.

Table 55.4 The output condition of VCOUT pin by the security setting

PORT Security Attribution (PmSAR (m = 0 to 9, A, B))	ACMPHSn Security Attribution (PSARm (m = B to E))	VCOUT pin output source
0 (secure)	0 (secure)	ACMPHSn output enable
0 (secure)	1 (non-secure)	ACMPHSn output disable
1 (non-secure)	0 (secure)	ACMPHSn output disable
1 (non-secure)	1 (non-secure)	ACMPHSn output enable

Note: n = 0 to 3

55.8 Usage Notes

55.8.1 Settings for the Module-Stop Function

ACMPHS operation can be disabled or enabled using the Module Stop Control Register. The ACMPHS is initially stopped after reset. Releasing the module-stop state enables access to the registers. For details, see section 11, Low Power Mode.

55.8.2 Settings for the DAC12

ACMPHS is connected to the internal module output of the D/A converter. For details, see section 53, 12-Bit D/A Converter (DAC12).

55.8.3 Relationship with the ADC16H

Constraints apply on the simultaneous use of ACMPHS analog input and ADC16H analog input. For details, see section section 52.11.10. Restrictions on Analog Channel Shared Among Multiple A/D Converters (ADC16H) and High-Speed Analog Comparator (ACMPHS).

55.8.4 ACMPHS Operation in Module-Stop State

When the module-stop state is entered while ACMPHS is operating, analog circuits in the ACMPHS are not stopped and the analog power supply current is the same as that when ACMPHS is being used. If the analog power supply current needs to be reduced in the module-stop state, set the CMPCTL.HCMPON bit to 0 to stop the ACMPHS.

55.8.5 ACMPHS Operation in Software Standby Mode

When Software Standby mode is entered while ACMPHS is operating, analog circuits in the ACMPHS are not stopped and the analog power supply current is the same as that when ACMPHS is being used. If the analog power supply current needs to be reduced in Software Standby mode, set the CMPCTL.HCMPON bit to 0 to stop the ACMPHS.

55.8.6 Setting the D/A Converter for Generating Reference Voltage

Set the D/A converter to generate reference voltage and wait for the D/A converter output settling time before enabling the comparator. Similarly, before making any changes to the settings of the D/A converter, stop the comparator temporarily. After the changes are made, wait for the D/A converter output settling time before enabling the comparator.